

EXPERIMENT-3

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USER INTERFACE AND DESIGN

Develop and compare CLI, GUI, and Voice User Interfaces (VUI) for the same task and assess user satisfaction.

1. COMMAND LINE INTERFACE

Python code:

```
import os

import sys

def rename_file(old_name, new_name):

    try:

        os.rename(old_name, new_name)

        print(f"File renamed from {old_name} to {new_name}")

    except FileNotFoundError:

        print(f"Error: {old_name} not found.")

    except Exception as e:

        print(f"An error occurred: {e}")

if __name__ == "__main__":

    if len(sys.argv) != 3:

        print("Usage: python rename_file_cli.py <old_filename> <new_filename>")

    else:

        rename_file(sys.argv[1], sys.argv[2])
```

Output:

```
PS E:\uid> python -u "e:\uid\cli.py"
● Usage: python cli.py <old_filename> <new_filename>
● PS E:\uid> python cli.py cli.py commandlineinterface.py
File renamed from 'cli.py' to 'commandlineinterface.py'
PS E:\uid> █
```

2. GRAPHICAL USER INTERFACE

```
import tkinter as tk

from tkinter import messagebox

import os

def rename_file():

    old_name = old_filename_entry.get().strip()

    new_name = new_filename_entry.get().strip()

    if not old_name or not new_name:

        messagebox.showwarning("Warning", "Please enter both filenames!")

    return

    if not os.path.exists(old_name):

        messagebox.showerror("Error", f"File '{old_name}' not found.")

    return

    if os.path.exists(new_name):

        if not messagebox.askyesno("Warning", f"'{new_name}' already exists. Overwrite?"):

            return

    try:
```

```

os.rename(old_name, new_name)

messagebox.showinfo("Success", f"File renamed from '{old_name}' to '{new_name}'")

except Exception as e:

messagebox.showerror("Error", f"An error occurred: {e}")

root = tk.Tk()

root.title("File Renamer")

root.geometry("400x150")

root.resizable(False, False)

tk.Label(root, text="Old Filename:", font=("Arial", 10)).grid(row=0, column=0, padx=10,
pady=5, sticky="w")

tk.Label(root, text=" New Filename:", font=("Arial", 10)).grid(row=1, column=0, padx=10,
pady=5, sticky="w")

old_filename_entry = tk.Entry(root, width=40)

old_filename_entry.grid(row=0, column=1, padx=10, pady=5)

new_filename_entry = tk.Entry(root, width=40)

new_filename_entry.grid(row=1, column=1, padx=10, pady=5)

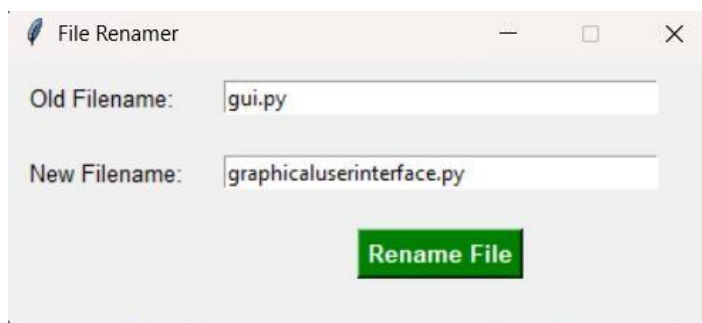
rename_button = tk.Button(root, text=" Rename File", bg="green", fg="white",
font=("Arial", 10, "bold"), command=rename_file)

rename_button.grid(row=2, column=0, columnspan=2, pady=10)

root.mainloop()

```

Output:





3. VOICE USER INTERFACE

Python Code:

```
import speech_recognition as sr

import os

def rename_file_from_voice_command(command):

    try:

        words = command.lower().split(" ")

        if "rename" in words and "to" in words:

            rename_index = words.index("rename")

            to_index = words.index("to")

            old_name = words[rename_index + 1]

            new_name = words[to_index + 1]

            if not os.path.exists(old_name):

                print(f"Error: File '{old_name}' not found.")

            return

            os.rename(old_name, new_name)

            print(f" File renamed from '{old_name}' to '{new_name}'")
```

```

except Exception as e:

print(f"Error: {e}")

def listen_for_command():

recognizer = sr.Recognizer()

mic = sr.Microphone()

print(" Listening for command to rename a file...")

with mic as source:

recognizer.adjust_for_ambient_noise(source)

audio = recognizer.listen(source)

try:

command = recognizer.recognize_google(audio)

print(f" Command received: {command}")

rename_file_from_voice_command(command)

except sr.UnknownValueError:

print("Sorry, I couldn't understand the command.")

except sr.RequestError as e:

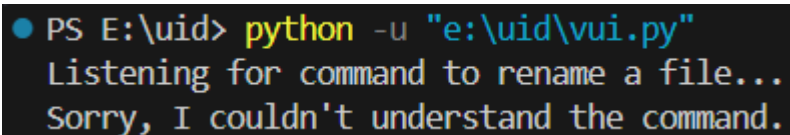
print(f" Could not request results from Google Speech Recognition service; {e}")

if __name__ == "__main__":

listen_for_command()

```

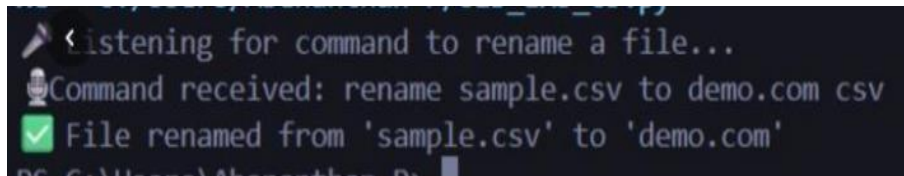
Output:



```

● PS E:\uid> python -u "e:\uid\vui.py"
Listening for command to rename a file...
Sorry, I couldn't understand the command.

```



4.USER SATISFACTION COMPARISON

Python Code:

```
def survey():

    print("Rate your satisfaction with the following interfaces (1-5):")

    try:

        cli_satisfaction=int(input("CLI(1-5):"))

        gui_satisfaction=int(input("GUI(1-5):"))

        vui_satisfaction=int(input("VUI(1-5):"))

        if not(1<=cli_satisfaction<=5 and 1<=gui_satisfaction<=5 and 1<=vui_satisfaction<=5):

            print("Please enter a number between 1 and 5 for each interface.")

            return

        print("\nYour satisfaction ratings:")

        print(f"CLI: {cli_satisfaction}")

        print(f"GUI: {gui_satisfaction}")

        print(f"VUI: {vui_satisfaction}")

        avg_satisfaction=(cli_satisfaction+gui_satisfaction+vui_satisfaction)/3

        print(f"Average Satisfaction: {avg_satisfaction:.2f}")

    except ValueError:

        print("Invalid input. Please enter a number between 1 and 5 for each interface.")

if __name__=="__main__":

    survey()
```

Output:

```
PS E:\uid> python -u "e:\uid\tempCodeRunnerFile.py"
Rate your satisfaction with the following interfaces (1-5):
CLI(1-5):3
GUI(1-5):4
VUI(1-5):2

Your satisfaction ratings:
CLI: 3
GUI: 4
VUI: 2
Average Satisfaction: 3.00
```

Result:

Thus , the Command Line Interface , Graphical User Interface , Voice Command Interface and User Satisfactory scores to rename a file is created and implemented successfully.